

Unified Chemical Synthesis of Corynanthe-Sarpagine-Ajmaline Type Monoterpene Indole Alkaloids: A Biomimetic Approach

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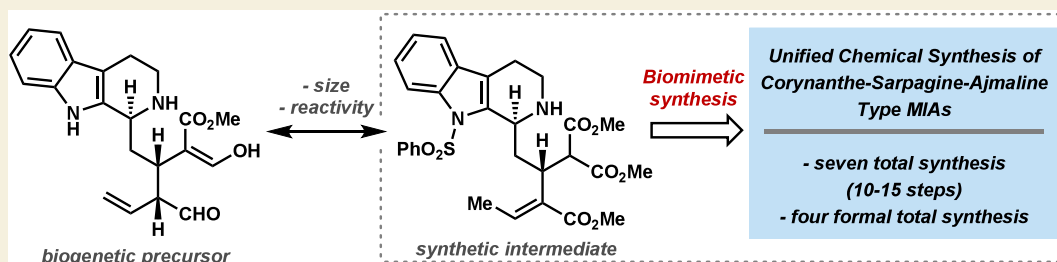
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ABSTRACT: The biosynthesis and chemical synthesis of corynanthe-sarpagine-ajmaline type monoterpene indole alkaloids, which share a close structural relationship, have presented key challenges in this field for decades. Herein, we present a biomimetic synthesis strategy inspired by biosynthetic hypotheses, successfully achieving a unified chemical synthesis of various corynanthe-sarpagine-ajmaline type polycyclic alkaloids. The success of this approach is attributed to (1) the rapid construction of chemically synthesized intermediates that closely resemble universally accepted biosynthetic intermediates via a sequence of formal Michael addition and *retro*-Michael addition reactions; (2) the efficient synthesis of the key indole-fused *aza*-bicyclic [3.3.1]nonane framework in the target molecules through regioselective oxidative dehydrogenation and intramolecular biomimetic Mannich cyclization for the first time; (3) leveraging TMSI-mediated intramolecular lactone cleavage/intramolecular N-alkylation cascade reaction to create the rigid 1-*aza* bicyclic [2.2.2]octane system in the target molecules, with selectivity controlled by the reaction solvent.

KEYWORDS: monoterpene indole alkaloids, biomimetic synthesis strategy, Mannich reaction, Michael addition

Traditionally, *Apocynaceae* plants have been utilized in folk medicine across Central America, Africa, and Asia for treating various diseases, including cancer, infections, and malaria, for centuries.¹ Sarpagine-ajmaline-type alkaloids are monoterpene indole alkaloids (MIAs) predominantly isolated from *Apocynaceae* plants, particularly from the genera *Alstonia* and *Rauwolfia*, with over 200 members reported (Figure 1). Preliminary pharmacological studies suggest that these MIAs play a crucial role in the diverse biological activities of *Apocynaceae* plants.^{2–4} Structurally, sarpagine alkaloids exhibit a distinctive polycyclic, cage-like architecture, encompassing an indole-fused *aza*-bicyclic [3.3.1]nonane framework (ABCE ring) and a rigid *aza*-bicyclic [2.2.2]octane framework (DE ring). Notable differences among these molecules arise from variations in stereochemistry at the C16 position and the configuration of the double bond at C19, contributing to significant variations in biological activities among family members. Ajmaline-type alkaloids are considered to be derivatives of sarpagine alkaloids, formed through C7–C17 bond formation, a transformation firmly established in their biosynthetic origin and a preferred strategy for the chemical synthesis of ajmaline-type alkaloids.⁵ Correspondingly, corynanthe-type alkaloids from the *Rubiaceae* family possess

relatively simpler polycyclic skeletal structures;⁶ however, these three types of MIAs share a strong biosynthetic connection.⁷

Due to their synthetically challenging frameworks and notable biological activities, especially those associated with sarpagine-type alkaloids, these molecules have garnered significant attention from organic chemists, leading to the successful total syntheses of several family members.⁸ These syntheses can generally be categorized into two types based on the sequence of indole ring formation (AB ring). The first type employs tryptophan or structural analogues containing the indole unit as starting materials, constructing the ABC tricyclic system initially, followed by the DE ring system. Notable examples include the Pictet–Spengler/Dieckmann cyclization method developed by the Cook group,^{9–15} the copper-catalyzed oxidative cyclopropanol ring-opening cyclization developed by the Zhang group,¹⁶ the innovative Mannich

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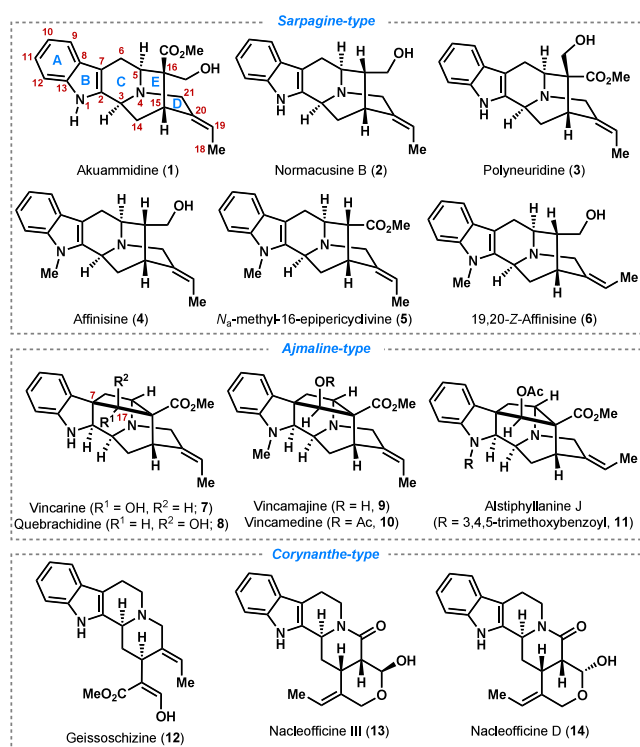


Figure 1. Representative corynanthe-sarpagine-ajmaline-type mono-terpenoid indole alkaloids.

cyclization/ SmI_2 -mediated coupling strategy reported by Zhang et al.,¹⁷ and the *aza*-Achmatowicz rearrangement/indole cyclization approach recently detailed by Qi and colleagues.¹⁸ The second type involves the initial construction of the CDE tricyclic system, followed by Fischer indole formation to complete the AB ring system. For instance, the Gaich lab utilized a [5 + 2] cycloaddition followed by palladium-catalyzed ketone α -alkylation and ring expansion, focusing on generating the CDE bicyclic skeleton before completing the synthesis of sarpagine alkaloids via Fischer indole formation.^{19,20} Similarly, the Takayama group identified gold(I)-catalyzed 6-*exo-dig* cyclization of alkynyl silyl enol ether as key step, first constructing the CDE tricycle and then forming the AB ring system through Fischer indole synthesis.²¹ More recently, the Tong group employed *aza*-Achmatowicz rearrangement/*aza*-Prins cyclization as critical steps, swiftly constructing the CDE tricyclic skeleton and subsequently synthesizing several sarpagine alkaloids through Fischer indole formation.²² Despite these advancements, the development of unified synthetic strategies, particularly those related to biosynthesis, for the family of corynanthe-sarpagine-ajmaline type MIAs remains crucial.

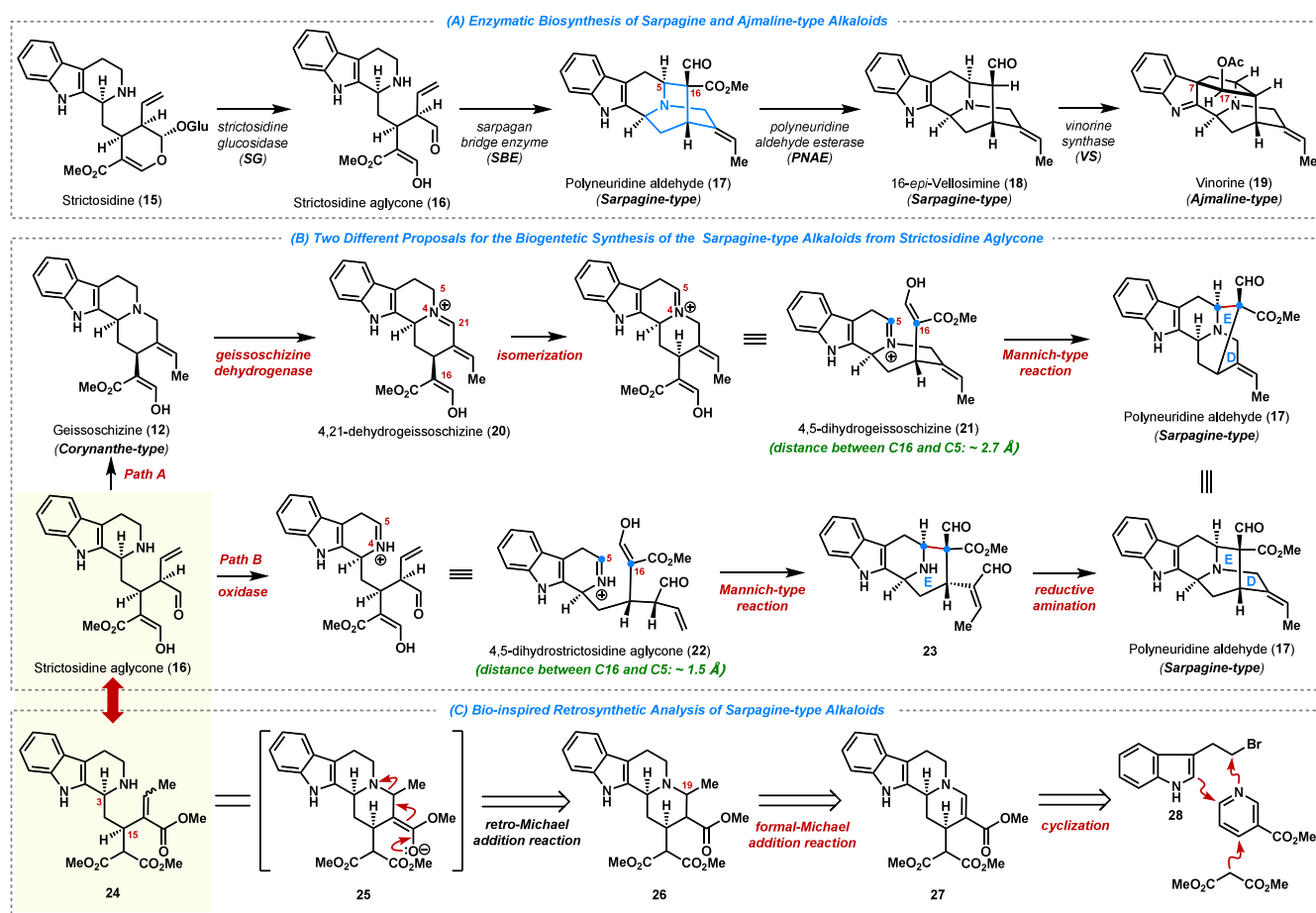
Much like chemical synthesis, the biosynthetic pathways of sarpagine-ajmaline-type alkaloids have attracted extensive investigation by natural product chemists.^{2,3,5,7,23} It has been established that, similar to the vast majority of MIAs (over 3000 members), sarpagine alkaloids originate from strictosidine (Scheme 1A). Through the action of strictosidine glucosidase (SG), strictosidine (15) is first converted to strictosidine aglycone (16), which is subsequently catalyzed by the sarpagan bridge enzyme (SBE) to yield the biosynthetic precursor polyneuridine aldehyde (17) via crucial C5–C16 bond formation. Following this, 17 undergoes a decarboxylation and intramolecular cyclization sequence—catalyzed by

polyneuridine aldehyde esterase (PNAE) and vinorine synthase (VS)—to produce the ajmaline-type alkaloid precursor vinorine (19). Overall, this complex reaction sequence comprises multiple catalytic steps involving various enzymes.

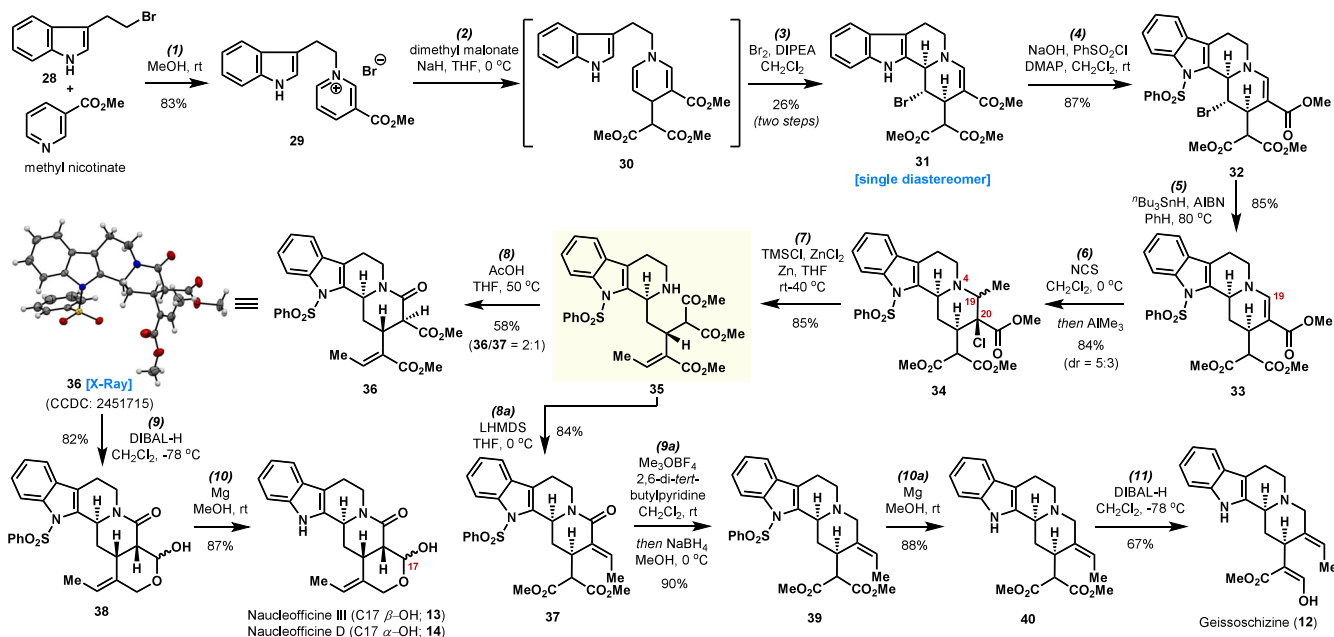
Although extensive studies have elucidated many details of the biosynthesis of these alkaloids, there is ongoing contention regarding the timing of C5–C16 bond formation that is critical for establishing the sarpagine framework. Currently, two competing hypotheses prevail (Scheme 1B). The first, proposed by van Tamelen in 1968,²⁴ posits that the strictosidine aglycone (16) is first converted into the corynanthe-type alkaloid Geissoschizine (12), which is then oxidized and isomerized to yield the imine ion 4,5-dihydro-geissoschizine (21). This culminates in an intramolecular Mannich reaction to afford polyneuridine aldehyde (17). Supporting this path, both van Tamelen's and Martin's groups have successfully synthesized sarpagine alkaloids or their structural analogues using various imine ion precursors.^{25–27} Most recently, O'Connor et al. identified that homologous cytochrome P450 enzymes from *Rauwolfia serpentina* and *Gelsemium sempervirens* could convert Geissoschizine (12) into 17 via an imine ion intermediate (like 21).²⁸ However, the detailed mechanism of enzymatic catalysis remains unclear, and this conversion is significantly dependent on the substrate. For substrates that lack suitable nucleophilic groups, the imine ion intermediate undergoes spontaneous oxidation, producing aromatic byproducts.²⁹ Moreover, their findings indicate that different cytochrome P450 enzymes could convert 12 into monoindole alkaloids (MIAs) with diverse skeletal structures.³⁰ Coincidentally, 25 years after van Tamelen proposed the aforementioned hypothesis, Lounasmaa reassessed this biosynthetic pathway. Although they successfully prepared an imine ion intermediate resembling 21 using the Polonovski–Potier approach,³¹ they did not observe the product of the intramolecular Mannich cyclization—a finding also encountered in Martin's synthetic studies.²⁷ Clearly, this biosynthetic hypothesis remains laden with uncertainties regarding both biological and chemical synthesis. In light of these findings, Lounasmaa proposed an alternative biosynthetic hypothesis, suggesting that the conformationally flexible imine ion 22 serves first to construct the E ring in the target molecules via an intramolecular Mannich reaction, followed by reductive amination for D ring construction, thereby facilitating the synthesis of the sarpagine alkaloid precursor polyneuridine aldehyde (17).^{4,32} Although this hypothesis lacks validation through either biological or chemical synthesis, the shortest possible distances between the reactive sites C5 and C16 in the imine ion intermediates 21 and 22 indicates that 22 may be more favorable for undergoing the intramolecular Mannich reaction, forging the critical C5–C16 bond.

In designing a unified chemical synthesis for corynanthe-sarpagine-ajmaline type MIAs, we were drawn to compound 24 (Scheme 1C), which exhibits a high structural similarity with the biosynthetic common intermediate, strictosidine aglycone (16). From both molecular framework and reactivity perspectives, 24 presented an ideal substitute for 16, with significant potential to mimic Lounasmaa's biosynthetic hypothesis, thus facilitating the biomimetic synthesis of sarpagine-ajmaline type alkaloids. For the synthesis of 24, we envisioned assembling it from tetracycle 26 via a *retro*-Michael addition reaction, in which 26 could be derived from 27 through the formal Michael addition of a methyl group.

Scheme 1. (A) Biosynthetic Pathway of Sarpagine-Ajmaline-Type Alkaloids, (B) Two Different Proposals for the Formation of 17 from 16, and (C) Our Bio-inspired Synthetic Design



Scheme 2. Synthesis of the Strictosidine Aglycone Substitute 35 and Its Transformation into the Corynanthe-Type MIAs



Compound **27** could be synthesized from the commercially available indole derivative **28**, methyl nicotinate, and dimethyl malonate.

Our synthesis commenced with the intermolecular alkylation of methyl nicotinate (Scheme 2). The resulting pyridine salt **29** reacted with dimethyl malonate to yield the unstable

Synthesis of 19,20-(Z)-Affinisine (6) and related compounds:

Reaction Scheme:

Starting material **35** (Phenylthio-substituted piperidine derivative) is converted to **41** (N-chloro intermediate) using (1) NCS, CH₂Cl₂, 0 °C (88% yield). **41** is then converted to **42** (proposed intermediate) using (2) NaH, CH₃CN/DCE, 40 °C (56% yield, entry 15 in Table 1). **42** is cyclized to **43** (lactone intermediate) using AlCl₃, 0 °C. **43** is then converted to **48** (lactone intermediate) using (3) DIBAL-H, CH₂Cl₂, -78 °C (68% yield). **48** is converted to **49** (lactone intermediate) using (4) LiOH, THF/H₂O, 55 °C (76% yield), then HOAc. **49** is converted to **51** (lactone intermediate) using (5a) CCl₄/MeOH, TMSI, 70 °C (76% yield). **51** is converted to **54** (lactone intermediate) using (7c) MeOH/THF, Mg, rt (93% yield). **54** is converted to **55** (lactone intermediate) using (8a) NaH, Me₂SO₄, THF, 0 °C, then LAH (75% yield). **55** is converted to **56** (lactone intermediate) using (7b) LAH, THF, 0 °C (88% yield). **56** is converted to **57** (lactone intermediate) using (7a) NaH, PMBCl, DMF, 0 °C (72% yield). **57** is converted to **58** (lactone intermediate) using (7b) LAH, THF, 0 °C (88% yield). **58** is converted to **59** (lactone intermediate) using (7a) NaH, PMBCl, DMF, 0 °C (72% yield). **59** is converted to **60** (lactone intermediate) using (7b) LAH, THF, 0 °C (88% yield). **60** is converted to **61** (lactone intermediate) using (7a) NaH, PMBCl, DMF, 0 °C (72% yield). **61** is converted to **62** (lactone intermediate) using (7b) LAH, THF, 0 °C (88% yield). **62** is converted to **63** (lactone intermediate) using (7a) NaH, PMBCl, DMF, 0 °C (72% yield). **63** is converted to **64** (lactone intermediate) using (7b) LAH, THF, 0 °C (88% yield). **64** is converted to **65** (lactone intermediate) using (7a) NaH, PMBCl, DMF, 0 °C (72% yield). **65** is converted to **66** (lactone intermediate) using (7b) LAH, THF, 0 °C (88% yield). **66** is converted to **67** (lactone intermediate) using (7a) NaH, PMBCl, DMF, 0 °C (72% yield). **67** is converted to **68** (lactone intermediate) using (7b) LAH, THF, 0 °C (88% yield). **68** is converted to **69** (lactone intermediate) using (7a) NaH, PMBCl, DMF, 0 °C (72% yield). **69** is converted to **70** (lactone intermediate) using (7b) LAH, THF, 0 °C (88% yield). **70** is converted to **71** (lactone intermediate) using (7a) NaH, PMBCl, DMF, 0 °C (72% yield). **71** is converted to **72** (lactone intermediate) using (7b) LAH, THF, 0 °C (88% yield). **72** is converted to **73** (lactone intermediate) using (7a) NaH, PMBCl, DMF, 0 °C (72% yield). **73** is converted to **74** (lactone intermediate) using (7b) LAH, THF, 0 °C (88% yield). **74** is converted to **75** (lactone intermediate) using (7a) NaH, PMBCl, DMF, 0 °C (72% yield). **75** is converted to **76** (lactone intermediate) using (7b) LAH, THF, 0 °C (88% yield). **76** is converted to **77** (lactone intermediate) using (7a) NaH, PMBCl, DMF, 0 °C (72% yield). **77** is converted to **78** (lactone intermediate) using (7b) LAH, THF, 0 °C (88% yield). **78** is converted to **79** (lactone intermediate) using (7a) NaH, PMBCl, DMF, 0 °C (72% yield). **79** is converted to **80** (lactone intermediate) using (7b) LAH, THF, 0 °C (88% yield). **80** is converted to **81** (lactone intermediate) using (7a) NaH, PMBCl, DMF, 0 °C (72% yield). **81** is converted to **82** (lactone intermediate) using (7b) LAH, THF, 0 °C (88% yield). **82** is converted to **83** (lactone intermediate) using (7a) NaH, PMBCl, DMF, 0 °C (72% yield). **83** is converted to **84** (lactone intermediate) using (7b) LAH, THF, 0 °C (88% yield). **84** is converted to **85** (lactone intermediate) using (7a) NaH, PMBCl, DMF, 0 °C (72% yield). **85** is converted to **86** (lactone intermediate) using (7b) LAH, THF, 0 °C (88% yield). **86** is converted to **87** (lactone intermediate) using (7a) NaH, PMBCl, DMF, 0 °C (72% yield). **87** is converted to **88** (lactone intermediate) using (7b) LAH, THF, 0 °C (88% yield). **88** is converted to **89** (lactone intermediate) using (7a) NaH, PMBCl, DMF, 0 °C (72% yield). **89** is converted to **90** (lactone intermediate) using (7b) LAH, THF, 0 °C (88% yield). **90** is converted to **91** (lactone intermediate) using (7a) NaH, PMBCl, DMF, 0 °C (72% yield). **91** is converted to **92** (lactone intermediate) using (7b) LAH, THF, 0 °C (88% yield). **92** is converted to **93** (lactone intermediate) using (7a) NaH, PMBCl, DMF, 0 °C (72% yield). **93** is converted to **94** (lactone intermediate) using (7b) LAH, THF, 0 °C (88% yield). **94** is converted to **95** (lactone intermediate) using (7a) NaH, PMBCl, DMF, 0 °C (72% yield). **95** is converted to **96** (lactone intermediate) using (7b) LAH, THF, 0 °C (88% yield). **96** is converted to **97** (lactone intermediate) using (7a) NaH, PMBCl, DMF, 0 °C (72% yield). **97** is converted to **98** (lactone intermediate) using (7b) LAH, THF, 0 °C (88% yield). **98** is converted to **99** (lactone intermediate) using (7a) NaH, PMBCl, DMF, 0 °C (72% yield). **99** is converted to **100** (lactone intermediate) using (7b) LAH, THF, 0 °C (88% yield). **100** is converted to **101** (lactone intermediate) using (7a) NaH, PMBCl, DMF, 0 °C (72% yield). **101** is converted to **102** (lactone intermediate) using (7b) LAH, THF, 0 °C (88% yield). **102** is converted to **103** (lactone intermediate) using (7a) NaH, PMBCl, DMF, 0 °C (72% yield). **103** is converted to **104** (lactone intermediate) using (7b) LAH, THF, 0 °C (88% yield). **104** is converted to **105** (lactone intermediate) using (7a) NaH, PMBCl, DMF, 0 °C (72% yield). **105** is converted to **106** (lactone intermediate) using (7b) LAH, THF, 0 °C (88% yield). **106** is converted to **107** (lactone intermediate) using (7a) NaH, PMBCl, DMF, 0 °C (72% yield). **107** is converted to **108** (lactone intermediate) using (7b) LAH, THF, 0 °C (88% yield). **108** is converted to **109** (lactone intermediate) using (7a) NaH, PMBCl, DMF, 0 °C (72% yield). **109** is converted to **110** (lactone intermediate) using (7b) LAH, THF, 0 °C (88% yield). **110** is converted to **111** (lactone intermediate) using (7a) NaH, PMBCl, DMF, 0 °C (72% yield). **111** is converted to **112** (lactone intermediate) using (7b) LAH, THF, 0 °C (88% yield). **112** is converted to **113** (lactone intermediate) using (7a) NaH, PMBCl, DMF, 0 °C (72% yield). **113** is converted to **114** (lactone intermediate) using (7b) LAH, THF, 0 °C (88% yield). **114** is converted to **115** (lactone intermediate) using (7a) NaH, PMBCl, DMF, 0 °C (72% yield). **115** is converted to **116** (lactone intermediate) using (7b) LAH, THF, 0 °C (88% yield). **116** is converted to **117** (lactone intermediate) using (7a) NaH, PMBCl, DMF, 0 °C (72% yield). **117** is converted to **118** (lactone intermediate) using (7b) LAH, THF, 0 °C (88% yield). **118** is converted to **119** (lactone intermediate) using (7a) NaH, PMBCl, DMF, 0 °C (72% yield). **119** is converted to **120** (lactone intermediate) using (7b) LAH, THF, 0 °C (88% yield). **120** is converted to **121** (lactone intermediate) using (7a) NaH, PMBCl, DMF, 0 °C (72% yield). **121** is converted to **122** (lactone intermediate) using (7b) LAH, THF, 0 °C (88% yield). **122** is converted to **123** (lactone intermediate) using (7a) NaH, PMBCl, DMF, 0 °C (72% yield). **123** is converted to **124** (lactone intermediate) using (7b) LAH, THF, 0 °C (88% yield). **124** is converted to **125** (lactone intermediate) using (7a) NaH, PMBCl, DMF, 0 °C (72% yield). **125** is converted to **126** (lactone intermediate) using (7b) LAH, THF, 0 °C (88% yield). **126** is converted to **127** (lactone intermediate) using (7a) NaH, PMBCl, DMF, 0 °C (72% yield). **127** is converted to **128** (lactone intermediate) using (7b) LAH, THF, 0 °C (88% yield). **128** is converted to **129** (lactone intermediate)

Entry	Conditions	Results
1	^t BuOK, THF, 0 °C	complex
2	KOH, MeOH, 0 °C → 55 °C	complex
3	KOH, MeCN, rt	35 (20%), 37 (44%)
4	KOH, THF, 40 °C	44 (35%), 46 (23%), 47 (15%)
5	LHMDS, THF, -78 °C → -20 °C	37 (60%)
6	LTMP, THF, -78 °C → -40 °C	37 (64%)
7	NaH, DMF, 0 °C	complex
8	NaH, DCE, rt → 60 °C	no reaction
9	NaH, 1,4-dioxane, rt → 55 °C	44 (81%)
10	NaH, THF, 0 °C → 40 °C; then NH ₄ Cl (aq.), rt	44 (30%), 45 (45%)
11	NaH, THF, 40 °C; then AlCl ₃ , THF, 0 °C	43 (45%), 44 (30%)
12	NaH, THF, 40 °C; then BF ₃ •OEt ₂ , THF, 0 °C	43 (16%), 44 (27%)
13	NaH, MeCN, 40 °C; then AlCl ₃ , MeCN, 0 °C	43 (43%), 37 (11%), 35 (23%)
14	NaH, MeCN/DCE (v/v = 1:1), 40 °C; then AlCl ₃ , MeCN, 0 °C	43 (76%), 35 (<5%)
15 ^b	NaH, MeCN/DCE (v/v = 1:1), 40 °C; then AlCl ₃ , MeCN, 0 °C	43 (56%), 35 (17%), 44 (8%)

35

37

44

45

46

47

NCS allowed for the conversion of **33** to the imine ion intermediate (not shown). The in situ addition of trimethyl aluminum resulted in a smooth nucleophilic addition of the methyl group to the imine ion, yielding **34** with a combined yield of 84% and a diastereomeric ratio of 5:3. After acquiring **34**, we proceeded to cleave the N4–C19 bond via *retro*-aza-Michael addition to synthesize strictosidine aglycone substitute **35**. We were pleased to find that treatment of both diastereoisomers of **34** with excess zinc powder under Lewis acid conditions (TMSCl/ZnCl₂) afforded the desired **35** in 85% yield as a single diastereomers.

After successfully obtaining the strictosidine aglycone substitute **35**, we explored the possibility of converting it into corynanthe-type MIAs³⁴ and quickly discovered that **35** could react in the presence of acetic acid to yield two separable lactams, **36** and **37**, with an excellent combined yield and 2:1 ratio. Notably, we acquired single crystals of **36**, which fully confirmed its relative configuration through X-ray diffraction. Correspondingly, **35** could be exclusively converted to **37** in the presence of non-nucleophilic strong bases like LHMDS. Subsequent reduction of **36** using DIBAL-H afforded a pair of diastereomers **38**, which were deprotected using the Mg/MeOH system, giving inseparable natural products Naucleoficine III (**13**) and D (**14**) with a combined yield of 87%, matching the spectral data with literature reports.^{35,36} Meanwhile, in the presence of Meerwein's salt, **37** was smoothly converted to the imine ion intermediate (not shown), which was subsequently reduced with sodium borohydride to give **39** in high yield (90%). After deprotecting the sulfonamide group in the Mg/MeOH system, the resulting **40** was reduced using DIBAL-H, achieving the synthesis of the representative corynanthe-type molecule Geissoschizine (**12**)^{37,38} in 59% yield over two steps.

After achieving the synthesis of several corynanthe-type MIAs, we turned our attention to the more synthetically challenging sarpagine-ajmaline-type alkaloids (Scheme 3). As discussed in our biomimetic synthesis design, we aimed to selectively convert **35** into imine intermediate **42**. This selective conversion presented a substantial hurdle due to the presence of the allylic hydrogen atom at the C3 position. Initial attempts at directly achieving the oxidative dehydrogenation of **35** to prepare imine **42**, employing high-valent iodine and peroxide, were unsuccessful. Fortunately, we found that utilizing NCS as an oxidizing agent could convert **35** into compound **41** in high yield (88%). Building on this result and considering the steric hindrance variations at the C3 and C5 positions, we examined the regioselective elimination of **41** under basic conditions in pursuit of obtaining the imine intermediate **42** (Table 1), which would subsequently undergo biomimetic intramolecular Mannich cyclization to construct the indole-fused *aza*-bicyclic [3.3.1]nonane framework in the target molecules. Initial trials with potassium *tert*-butoxide and potassium hydroxide as bases resulted in only the decomposition of substrate **41** (entries 1–2). Interestingly, modifying the solvent in the potassium hydroxide system from methanol to acetonitrile or tetrahydrofuran (THF) gave completely different results (entries 3–4). In particular, in the THF system, although we did not obtain the desired cyclization product **43**, we observed the formation of lactams **46** and **47** (entry 4). We hypothesized that these lactams likely arose from the tautomerization of imine intermediate **42**, followed by selective lactamization. Encouraged by these findings, we explored non-nucleophilic strong bases; however, under LHMDS and LTMP conditions, we obtained lactam **37** in only good yield (entries 5–6). Further investigation revealed that employing sodium hydride as the base under different solvent conditions yielded markedly different reactivities for substrate **41** (entries 7–10). For example, in the 1,4-dioxane system, the C4–C16-bonded product **44** was produced in high yield (entry 9). However, under THF conditions, in addition to **44**, a more polar intermediate formed, which was completely converted to aromatic compound **45** when the reaction was quenched with an ammonium chloride aqueous solution (entry 10). We attempted to separate this unstable

intermediate but were unsuccessful. Although we have not characterized this reaction intermediate, the formation of **45**–**47** leads us to believe that it may represent imine compound **42**. Given this understanding, we sought to activate **42** in situ with a Lewis acid to promote further intramolecular Mannich cyclization. Fortunately, we discovered that adding the reaction mixture of **41** to a solution of aluminum chloride in THF successfully afforded the desired cyclization product **43** in 45% isolated yield (entry 11). The use of boron trifluoride etherate as the activating Lewis acid could also yield **43**, albeit with less favorable reaction outcomes (entry 12). Ultimately, we re-examined the reaction solvent using sodium hydride as the base and found that a mixed solvent system of acetonitrile and 1,2-dichloroethane (DCE) could produce the desired **43** in 76% yield, alongside a small amount of compound **35** (entry 14). When we applied these reaction conditions on a scale of 200 mg, the amount of **35** significantly increased (up to 17%); however, we were still able to obtain the desired product **43** with a 56% isolated yield (entry 15). Notably, **35** can be efficiently converted into cyclization precursor **41** in a single step, thereby minimizing the waste of starting materials and providing a strong basis for the synthesis of subsequent target molecules.

Following the successful biomimetic Mannich cyclization reaction, we redirected our efforts to construct the D ring of the target molecules (Scheme 3). To this end, the selective reduction of **43** with DIBAL-H caused the resulting allylic alcohol to spontaneously undergo intramolecular lactonization to give lactone **48** in 68% yield. We then employed lithium hydroxide for hydrolysis and decarboxylation, followed by relactonization in the presence of acetic acid to give lactone **49** in 76% yield. With **49** in hand, initial attempts to directly perform intramolecular N4 alkylation under basic conditions for forming the N4–C21 bond proved to be unsuccessful. However, we found that treating **49** with TMSI in CHCl₃/MeOH mixed solvent under heating conditions afforded the desired cyclization product **50**, along with the formation of its stereoisomer **51** in a combined yield of 51% and a 1:1.4 ratio (see the table with a gray background in Scheme 3, entry 1). Encouragingly, reducing the reaction temperature significantly enhanced the selectivity, leading to the formation of **50** in 78% yield (entry 2). Replacing CHCl₃ with DCE maintained a high selectivity but significantly slowed the reaction (entry 3). Further screening revealed that using nonpolar solvents, such as CCl₄ or PhMe (entries 4–5), completely inverted the reaction's selectivity, yielding **51** with a single *Z*-configuration of the double bond at the C19 position. Mechanistically, we speculate that iodide ions serve as nucleophiles, engaging in an intermolecular S_N2' ring-opening reaction with the TMS-activated allylic alcohol ester, followed by N4 acting as a nucleophile performing an intramolecular S_N2' cyclization with the in situ generated allylic iodide (for the details, see Supporting Information). This process facilitates the inversion of the double bond's geometric configuration. Nonetheless, this solvent-controlled selectivity allows us to conveniently obtain cyclization products with different double bond configurations, establishing a solid foundation for synthesizing diverse sarpagine-ajmaline-type alkaloids. For instance, the removal of the sulfonyl protecting group at the N1 position in **50**, followed by methylation, enabled the total synthesis of the sarpagine-type natural product N_a-methyl-16-epipericyclivine (**5**)³⁹ in 74% yield over two steps. Further reduction of **5** led to the synthesis of Affinisine (**4**)⁴⁰ in high yield (90%).

Meanwhile, the simple reduction of **52** also facilitated the total synthesis of Normacusine B (**2**).⁴¹ By protecting the N1 atom in **52** with a PMB group, compound **53** was obtained in high yield, which is a known advanced intermediate for the synthesis of several sarpagine-ajmaline-type MIAs,¹⁷ thereby achieving their formal total synthesis. Through similar transformations, compound **51** could also undergo deprotection, methylation, and one-pot reduction of the ester group to facilitate the total synthesis of the natural product 19,20-Z-afnisine (**6**)⁴² in 70% overall yield, which has a Z-configuration of the double bond at the C19 position. It should be noted that the spectral properties of our synthetic natural products and advanced intermediate were in good agreement with those reported in the literatures.

In conclusion, inspired by Lounasmaa's biosynthetic hypothesis, we have successfully validated this hypothesis through chemical synthesis for the first time. Furthermore, we have achieved a unified chemical synthesis of seven corynanthe-sarpagine-type MIAs through 10 to 15 steps of chemical transformations from commercially available materials, along with the formal total synthesis of four sarpagine-ajmaline-type alkaloids. We believe that the synthetic strategy demonstrated here not only provides essential synthetic chemical support for the biosynthesis of corynanthe-sarpagine-ajmaline type MIAs but also offers a valuable platform for the chemical synthesis of related alkaloids and their structural analogues, thereby facilitating further exploration of their biological activities.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/jacsau.5c00903>.

Experimental procedures and spectroscopic data (PDF)

Accession Codes

Deposition Number 2451715 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge via the joint Cambridge Crystallographic Data Centre (CCDC) and Fachinformationszentrum Karlsruhe Access Structures service.

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Author Contributions

Yu-Ming Zhao conceived the work and designed the experiments. Ya-Kui Sun, Xiao-Feng Guo, and Yu Bu performed the experiments and analyzed the data. Ya-Kui Sun and Jin-Bao Qiao prepared the Supporting Information. Yu-Ming Zhao guided the project and wrote the manuscript. All the authors checked the manuscript.

Notes

The authors declare no competing financial interest.

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